MERN STACK

TRAINING TR-103 REPORT DAY 14

10 July 2026

Overview:

The fourteenth day of the MERN Stack training focused on working with time and dates in JavaScript. We learned how to use `setInterval()` to execute code repeatedly after a fixed time interval and `clearInterval()` to stop the repeated execution. We also explored the Date object and its methods for creating and manipulating dates and times. These concepts were applied by developing a Stopwatch and a Digital Clock, helping us understand real-time JavaScript applications.

Learning Objectives:

- Learn how to stop repeated execution using `clearInterval()`.
- Understand the `setInterval()` method.
- Explore the JavaScript `Date` object.
- Learn commonly used Date methods.
- Build a Stopwatch and Digital Clock using JavaScript.

JAVASCRIPT

setInterval()

The setInterval() method is used to execute a function repeatedly after a specified time interval (in milliseconds). It continues running until it is stopped using clearInterval().

Features:

- Executes a function repeatedly.
- Uses time intervals in milliseconds.
- Commonly used for clocks, timers, and animations.

Syntax:

```
let intervalId = setInterval(function, delay);
```

clearInterval()

The clearInterval() method stops the repeated execution started by setInterval().

Features:

- Stops a running interval.
- Prevents continuous execution.
- Used with timers and stopwatches.

Syntax:

```
clearInterval(intervalId);
```

Date Object

The Date object is used to create, display, and manipulate dates and times in JavaScript. It can retrieve the current date and time or create a specific date.

Common Date Methods:

- getFullYear() – Returns the year.

- getMonth() – Returns the month.
- getDate() – Returns the day of the month.
- getDay() – Returns the day of the week.
- getHours() – Returns the hours.
- getMinutes() – Returns the minutes.
- getSeconds() – Returns the seconds.
- getTime() – Returns the time in milliseconds.

Stopwatch (Task 1)

A Stopwatch was developed using JavaScript by combining setInterval() and clearInterval(). It can start, stop, and reset time, demonstrating real-time event handling.

Features:

- Start timer.
- Stop timer.
- Reset timer.
- Display elapsed time.

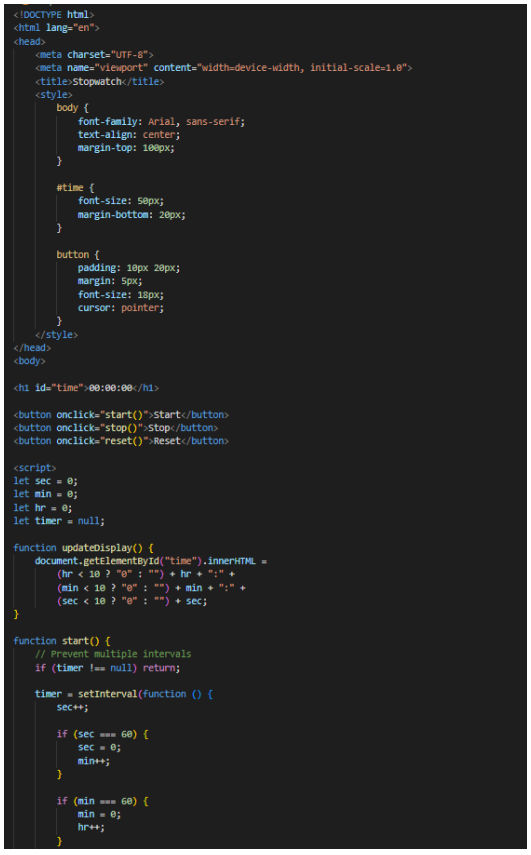
Digital Clock (Task 2)

A Digital Clock was created using the Date object and setInterval() to display the current time and update it every second.

Features:

- Displays current time.
- Updates automatically every second.
- Uses Date methods to retrieve hours, minutes, and seconds.

TASK :



```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Stopwatch</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            text-align: center;
            margin-top: 100px;
        }

        #time {
            font-size: 50px;
            margin-bottom: 20px;
        }

        button {
            padding: 10px 20px;
            margin: 5px;
            font-size: 18px;
            cursor: pointer;
        }
    </style>
</head>
<body>

<h1 id="time">00:00:00</h1>

<button onclick="start()">Start</button>
<button onclick="stop()">Stop</button>
<button onclick="reset()">Reset</button>

<script>
let sec = 0;
let min = 0;
let hr = 0;
let timer = null;

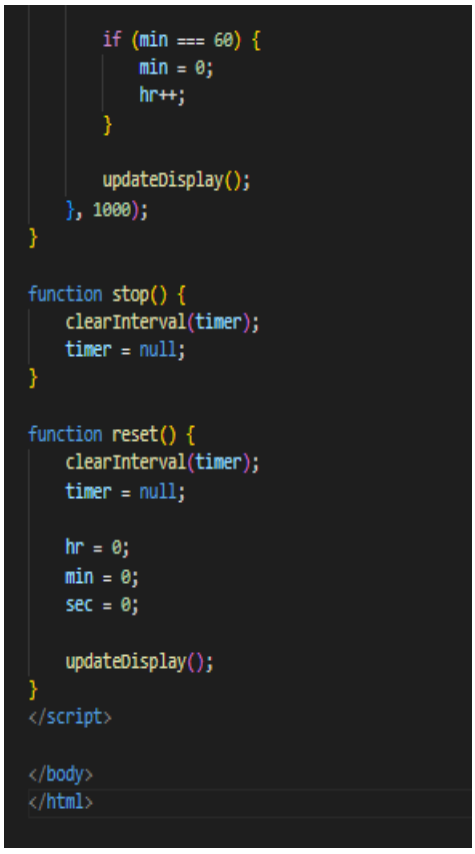
function updateDisplay() {
    document.getElementById("time").innerHTML =
        (hr < 10 ? "0" : "") + hr + ":" +
        (min < 10 ? "0" : "") + min + ":" +
        (sec < 10 ? "0" : "") + sec;
}

function start() {
    // Prevent multiple intervals
    if (timer !== null) return;

    timer = setInterval(function () {
        sec++;

        if (sec === 60) {
            sec = 0;
            min++;
        }

        if (min === 60) {
            min = 0;
            hr++;
        }
```



```
        if (min === 60) {
            min = 0;
            hr++;
        }

        updateDisplay();
    }, 1000);
}

function stop() {
    clearInterval(timer);
    timer = null;
}

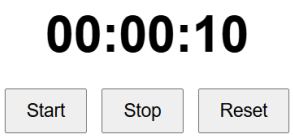
function reset() {
    clearInterval(timer);
    timer = null;

    hr = 0;
    min = 0;
    sec = 0;

    updateDisplay();
}
</script>

</body>
</html>
```

Output :



Conclusion:

By the end of Day 14, I learned how to work with time-based functions in JavaScript using setInterval(), clearInterval(), and the Date object. I applied these concepts by creating a Stopwatch and a Digital Clock, improving my understanding of real-time web applications.